WARIT YUVANIYAMA (NOTE)

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Fourth Year Computer Engineering Student

Sirindhorn International Institute of Technology (SIIT), Thammasat University

SUMMARY

High-achieving fourth-year Computer Engineering student (3.92 CGPA) and President of SIIT Robotic Club (OrcaBOT). Proficient in **embedded systems design** using **C++, Python, and ROS2**, with peer-reviewed research publications. Proven leader in managing technical procurement (budgeting **100,000+ THB**) and an award-winning communicator with a track record in technical pitching for innovative solutions in EdTech and Sustainability.

EDUCATION

Bachelor of Computer Engineering, SIIT, Thammasat University — Expected 2027

CGPA: 3.92 (Expected First-Class Honors); Recipient of Academic Outstanding Student Scholarship.

- *Relevant Coursework:* Machine Learning & Pattern Recognition, Natural Language Processing, Big Data Analytics & Machine Learning, Computer Networks, Computer Vision, Quantum Computing, Operating Systems, Database Systems, Algorithm Design.

High School (DPSTE Program), Samsenwittayalai School — Graduated Feb 2023

CGPA: 3.96

- *Relevant Coursework:* Computer and Algorithm, Programming, Introduction to Research Methodology, Design and Technology, Science Project, English for Academic Purpose.

SKILLS

Programming	Python, C++, Java, ROS2, micro-ROS, Basic Assembly, LaTeX
Systems & DB	Relational Database Design, SQL (MySQL), Ubuntu Linux
Hardware	ESP32, Raspberry Pi, Arduino, KiCad, Sensors/Actuators, MATLAB
Technical	Project Management, Research & Technical Writing, Technical Pitching
Soft Skills	System Thinking, Creative Problem Solving, Leadership, Cross-cultural Teamwork
Languages	Thai (Native), English (Fluent)
English Scores	TU-GET: 850 — SAT: 1420 (M:790, EBRW:630) — TGAT1: 96.7 — TOEFL (ITP): 553 (B2)

EXPERIENCE

President & Embedded Systems Lead, SIIT Robotic Club (OrcaBOT) — Oct 2023 – Present

- **Leadership:** Leading a 30-member club, managing a **100,000+ THB** procurement budget, and competing in **RoboCup Japan Open 2025** with an automatic soccer robot.
- **Systems Integration:** Orchestrating cross-departmental collaboration (High-level, Embedded, Electrical, Mechanical) to ensure seamless hardware-software integration for automatic soccer robots.
- **Firmware Architecture:** Developed custom C++ libraries for motor control and implemented holonomic drive algorithms; redesigned serial and wireless nRF radio communication systems to significantly increase responsiveness and stability.
- **Technical Communication:** Primary author for the Software Architecture section (System Overview, Low-Level, and High-Level Design) of the RoboCup 2024 Team Description Paper, translating complex cross-departmental logic into a peer-reviewed format. [Link: Team Description Paper (RoboCup 2024)]

Research Intern, Measurement & Intelligence Systems Lab, Kansai University, Japan — Jun–Jul 2026

- Developed a **mmWave radar + thermal imaging sensor fusion** pipeline for autonomous robot navigation in smoke-filled, low-visibility environments where RGB cameras and LiDAR degrade (TI IWR1443 radar, FLIR Boson 320 LWIR on TurtleBot3).
- Built the full **ROS2 Humble** integration: custom sensor drivers and UDEV workflow, hardware-accelerated JPEG streaming of 16-bit radiometric thermal frames, TF-tree sensor extrinsics, camera/radar intrinsics calibration, and **ApproximateTimeSynchronizer**-based multi-sensor fusion overlaying depth-colored radar point clouds in real time.

- Engineered WSL2/WSL networking (FastDDS whitelisting, kernel buffer tuning, UDEV persistent port naming) to sustain high-frequency sensor streams between the Raspberry Pi 3B+ robot and remote CUDA-enabled PC.

Intern, NECTEC, NSTDA May 2024 – Dec 2025

- Implemented Fast Segment Anything Model (FastSAM) for water surface level monitoring system.
- Developed a scaled wave-tank coastal erosion model to test the Water Surface Level Estimation model.
- Presented research findings at international venues, including IUKM 2025 (Vietnam) and the International Conference on Biodiversity 2025 (IBD2025, Queen Sirikit National Convention Center, Bangkok). Published in Springer LNCS/LNAI vol 15585.

Intern, Chemistry Department, Faculty of Science, Mahidol University Mar 2022 (10 days)

- Assisted in organic chemistry synthesis; achieved a 33.8% yield (2.35g) for naphtho[2,1-b]furan-1(2H)-one.
- Utilized ChemDraw software and performed laboratory troubleshooting.

RESEARCH PUBLICATIONS

- Chetsawang, J., Sricharoenchit, K., Jairak, K., **Yuvaniyama, W.** ***(Presenter)***, *et al.* (2025). Water Surface Level Estimation Based on the Fast Segment Anything Model for Coastal Monitoring Systems. In: Huynh, V.N., et al. (eds) *Integrated Uncertainty in Knowledge Modelling and Decision Making. IUKM 2025.* Lecture Notes in Computer Science, vol 15585. Springer, Singapore. DOI: 10.1007/978-981-96-4606-7_28
- **Yuvaniyama, W.** ***(First Author)***, Phinyachok, A., Sanguankittipun, G., Colombo, L., Boonkwan, P. (2026). Thai Poetry Rhyme Verification Using a Hybrid Rule-Based Approach for Education and GenAI. *iSAI-NLP 2026* (In review; submitted Sep 2026). *Improved rule-based rhyme verifier merged into PyThaiNLP and a dictionary-enhanced checker evaluated on 36,475 classical stanzas; serves as a deterministic reward signal for GenAI-based Klon-Paed generation.*
- Chetsawang, J., **Yuvaniyama, W.**, Boonkwan, P. (2027). Direction-Aware Reward Shaping for Fuel-Efficient Orbital Transfer using Reinforcement Learning. *IEICE Transactions on Communications, vol. E110-B, no. 6* (Special Section on Space, Aeronautical and Navigation Electronics, ICSANE2025). *Physics-informed PPO reward shaping via Jacobian linearization, reducing peak fuel expenditure by ∼10%.*
- Chetsawang, J., **Yuvaniyama, W.**, Jairak, K., Sricharoenchit, K., Leung, B.G., Mae, Y., Ratsamee, P. (2027). Online Rapid Innate Learning via Recursive Least-Squares for Online Legged Locomotion Learning. *IEEE/SICE International Symposium on System Integration (SII 2027)*, Kobe, Japan. *Online RLS refinement of legged-locomotion controller weights, reducing heading error from 26.3° to 5.5°; hardware feasibility on Unitree Go2.*
- Thongsupan, T., Sureechainirun, N., **Yuvaniyama, W.**, *et al.* (2024). OrcaBOT Team Description. *RoboCup 2024 Symposium and World Championship: Team Description Papers - Small Size League.* [Semantic Scholar]

PROJECTS

- **FlashLight: Companion Study Pacer Application** *Co-Founder, Operations Lead & Pitcher*
  Co-founded FlashLight (flashlight.in.th), a study-planning platform that helps Thai students pace their preparation for the high-stakes TCAS university admission exam by breaking 368+ A-Level topics into personalized daily missions with progress tracking and coaching. Delivered the pitch in every elimination round; reached the **Top 5** (of 80+ teams) with an iOS MVP that ranked **#7 in Education** on the Thai App Store.
- **CareAir: Air Quality Monitoring & Carbon Capture System (Hackathon Winner)**
  Developed "CareAir", a proof-of-concept carbon capture machine designed to improve air quality in classrooms/workplaces. Awarded **1st Place** (Physical Wellness Track) at Thammasat Hackathon: Future Wellness 2024.
- **CrystalEyes: Automated Crystal Imaging System**
  Developed a low-cost (10k-15k THB) CNC-based digital microscope system to automate crystallization screening, replicating core functionality of commercial systems costing 1M+ THB. Iteratively designed and built from high school through university. Demonstrated with lysozyme growth and MOF studies.
  Links: (Video) — (Report/Poster)
- **Autolocate: Condo Parking Management System**
  Designed a relational database backend using **SQL and Express.js** to manage high-traffic condo parking operations, implementing a three-tier connection pool system to enforce role-based access control (RBAC) and ensuring secure authentication via JWT and HttpOnly cookies.
  Link: (GitHub Repository)

- **Bedroom Tree: Enzymatic Indoor Carbon Capture Concept**
  Co-developed concept for an indoor CO_2 capture device using carbonic anhydrase enzyme. Reached semi-final round, Thailand Innovation Award 2023.
  Link: (Report/Poster)
- **Thailand CANSAT - Rocket Competition 2022**
  Designed mission parameters and internal satellite subsystems for the team's CANSAT submission.

WORKSHOPS & INTERNATIONAL PROGRAMS

- **Staff, PBL Workshop: Automatic Robot in Industrial Conveyor Belt Challenge** August 2026
  Planned and organized the event, built the TA demonstration robot, and supervised each participant team. Technical theme: multi-sensor navigation using **IMU, wheel encoders, and ultrasonic sensors with PID control** to identify robot heading and enable industrial conveyor-belt navigation.
- **Staff, International Project-Based Learning (iPBL) Workshop: Rescue Robot** August 2025
  Managed technical logistics and material procurement for international teams from SIIT and OIT Japan. Provided hands-on mentorship in robot design and system debugging, specializing in ROS2, micro-ROS, and Linux integration on Raspberry Pi and ESP32 platforms, while facilitating cross-cultural collaboration between Thai and Japanese students.
- **Participant, 11th Problem-Based Learning (PBL) Workshop, Taipei Tech, Taiwan** August 2024
  Collaborated within an international team (12 universities represented) to design and build a smart autonomous vehicle using Arduino, sensors/actuators, and 3D modeling for tasks including color recognition, positioning, and object grasping over an 8-day period.
- **Participant, OIT-SIIT International Project-Based Learning (iPBL), Osaka, Japan** July 2024
  Developed a functional mini rescue robot prototype from scratch in 1 week using ROS (on Raspberry Pi) for communication/video streaming and Arduino for motor/arm control; successfully completed simulated rescue mission objectives.

AWARDS AND RECOGNITION

- **3rd Runner-Up (Top 5)**, Chulalongkorn University Startup Thailand League (CUSTL) — FlashLight Team
  Selected as an **official representative of Chulalongkorn University** for the Startup Thailand League 2026 **Regional Pitching Round (Central Thailand)**.
- **Qualified & Exhibited**, SITE2026 Demo Day (NIA, Paragon Hall) — Startup Thailand League 2026
  Pitched FlashLight to VCs and investors from Thailand, South Korea, and beyond.
- **1st Place Winner**, Thammasat Hackathon: Future Wellness 2024 (CareAir Team)
- **Finalist**, ASEAN-China-India Youth Leadership Summit 2024 Sustainability Startathon (Singapore)
- **3rd Place (of 200 teams)**, 5th Kibo Robot Programming Challenge Thailand (Plumber Pups Team)
  Developed autonomous navigation and **ArUco marker detection** scripts in **Java** for ISS Astrobee robots.
- **Semi-finalist**, Thailand Innovation Award 2023 (Bedroom Tree Project)
- **1st Place (Gold Medal)**, Science Project & Invention, SKIT National Competition 2023
- **Research & Science Project Awards**, 2022 DPSTE Conference (1st Place Poster, 2nd Place Oral)